

TPG4190 Seismic data acquisition and processing

Lecture 6: Processing

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Overview

- ▶ Basic processing sequence
- ▶ Velocity analysis
- ▶ Migration

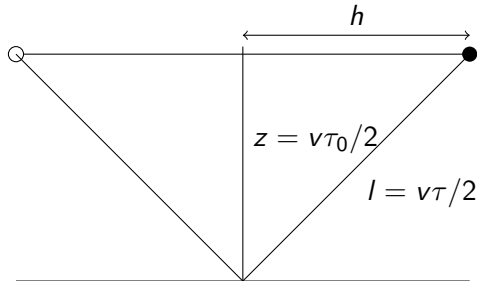
Basic processing sequence

1. Input data
2. Velocity analysis
3. NMO + Stack
4. Output result

Basic+ processing sequence

1. Input data
2. Velocity analysis
3. Migration
4. Output result

NMO and Stack



The travelttime $\tau(h)$ is:

$$l^2 = z^2 + h^2 \tag{1}$$

which gives by inserting $v\tau/2$ for l and $v\tau_0/2$ for z

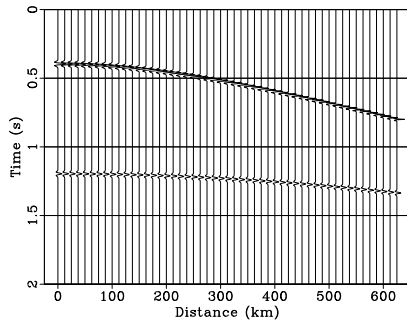
$$\tau(h) = \sqrt{\tau_0^2 + 4h^2/v^2}. \tag{2}$$

NMO and Stack

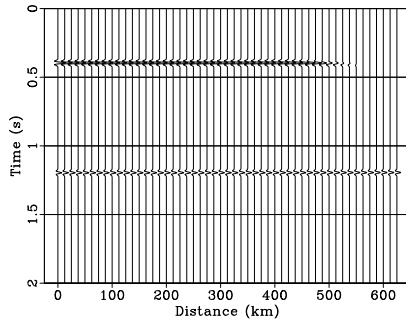
Nmo-correction:

$$\Delta\tau = \tau(h) - \tau_0, \quad (3)$$

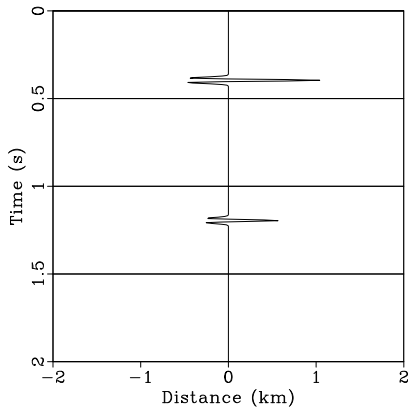
Cmp



Nmo



Stack



NMO and Stack

Average velocity v_{rms} defined by

$$v_{rms}^2(t_0) = \frac{1}{t_0} \int_0^{t_0} v^2(t) dt \quad (4)$$

$v(t)$: Interval velocity. The travelttime equation (2) then becomes

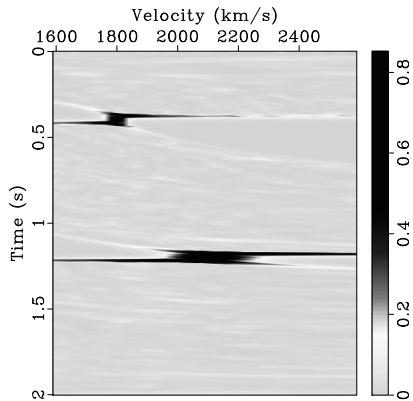
$$\tau(h) = \sqrt{t_0^2 + 4h^2/v_{rms}^2(t_0)}. \quad (5)$$

Velocity analysis

Estimate v_{rms} from equation (2)

1. Make a guess of v
2. Perform Nmo correction for all t_0 using guess of v
3. Make a stack trace
4. Perform above steps for a range of guesses of v
5. Plot all stack traces in a velocity spectrum

Velocity spectrum



Semblance

Stack is usually replaced with semblance to get better velocity spectra

$$S(t) = \frac{\int_0^{h_{max}} dh p^2(t, h)}{\int_0^T dt \int_0^{h_{max}} dh p^2(t, h)} \quad (6)$$

- ▶ $p(t, h)$: data
- ▶ h_{max} : maximum offset.

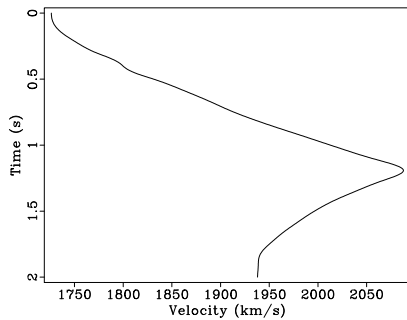
This equation can not be used directly. Why?

Semblance

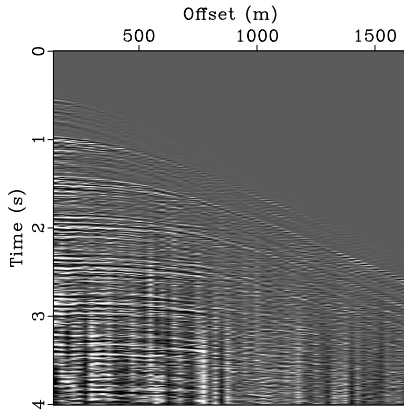
$$S_k = \frac{\sum_{i=0}^{N_h} p_{k,i}^2}{\sum_{k=0}^{N_t} \sum_{i=0}^{N_h} p_{k,i}^2} \quad (7)$$

- ▶ $S_k = S(t = k\Delta t), k = 0, \dots, N_t$
- ▶ $p_{k,i} = p(t = k\Delta t, h = i\Delta h), i = 0, \dots, N_h.$
- ▶ Δt time sampling interval,
- ▶ Δh distance between offsets
- ▶ N_t, N_h : No of time samples and No of offsets

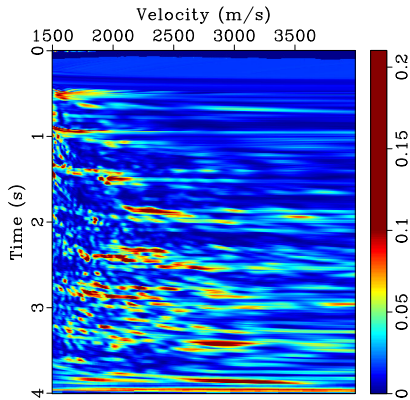
Velocity analysis



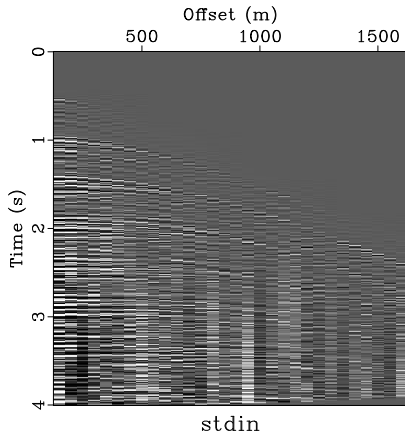
Velocity analysis



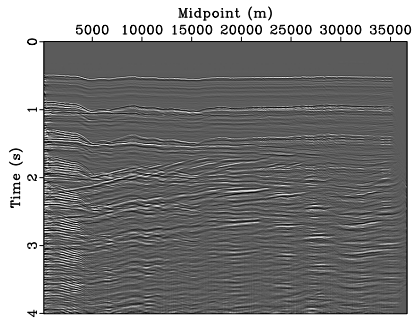
Velocity analysis



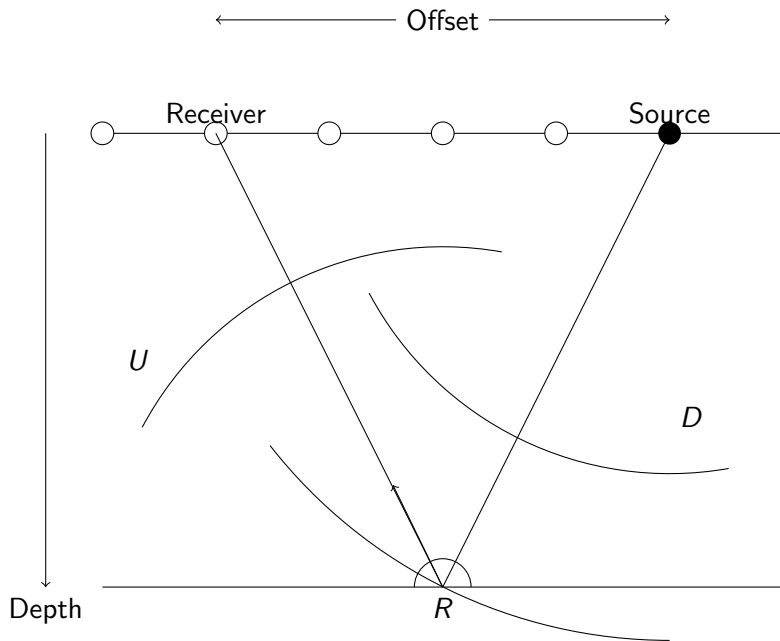
Velocity analysis



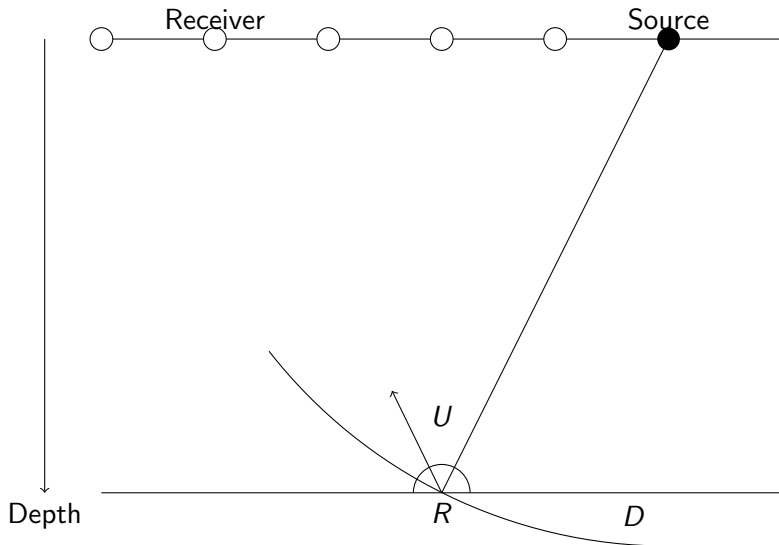
Velocity analysis



Migration



Migration principles



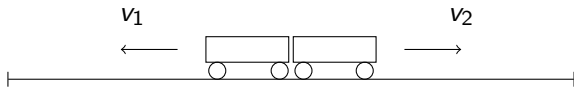
Migration principles

Make an image by cross-correlating the downgoing and upgoing waves

$$R(\mathbf{x}) = \int dt U(\mathbf{x}, t) D(\mathbf{x}, t), \quad (8)$$

where R is the reflectivity and $\mathbf{x} = (x, y, z)$ denotes a position in space, while t is the time.

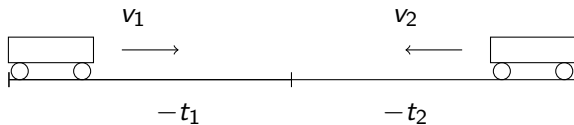
Time reversal 1



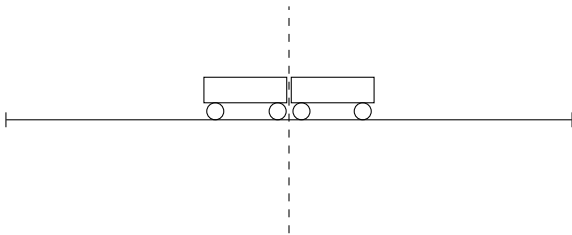
Time reversal 2



Time reversal 3



Time reversal



Seismic imaging

Forward modeling

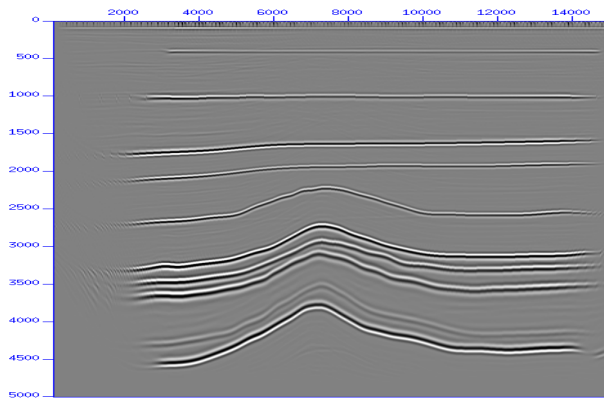
Time reverse modeling

Backward modeling

Cross correlation

Imaging

Complete image



Kirchhoff migration

Simplest approach for up- and downgoing waves

$$\begin{aligned}D(\mathbf{x}, t) &= A\delta(t - \tau_s), \\U(\mathbf{x}, t) &= BP(t + \tau_r),\end{aligned}\tag{9}$$

where

- ▶ $\mathbf{x}$: reflection point,
- ▶ A and B : amplitude factors
- ▶ P : Data at the surface
- ▶ τ_s travelttime from the source to the reflection point
- ▶ τ_r travelttime from the receiver to the reflection point

Kirchhoff migration

Using equation (9) and equation (8), I get

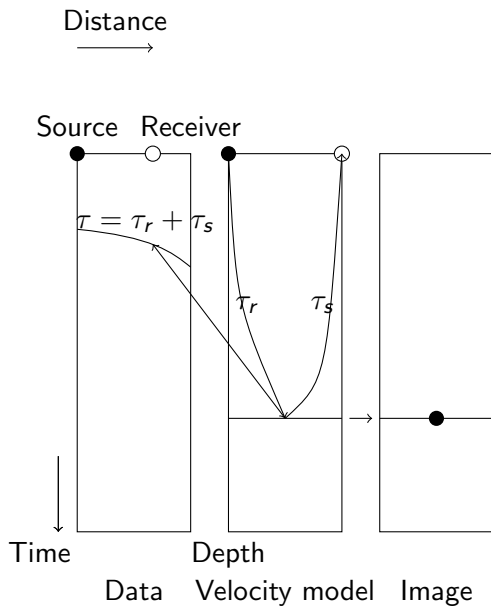
$$R(\mathbf{x}) = \int dt A\delta(t - \tau_s)BP(t + \tau_r), \quad (10)$$

which after integration gives

$$R(\mathbf{x}) = ABP(\tau_s + \tau_r). \quad (11)$$

If we disregard the amplitude factor AB the image is simply equal to the amplitude of the recorded data at a time equal to $\tau = \tau_r + \tau_s$.

Kirchhoff migration



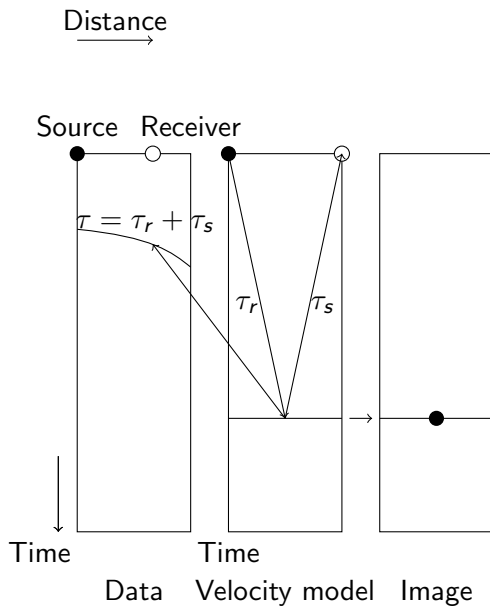
Kirchhoff migration

Many source-receiver pairs will contribute to the same imaging point, so that equation (11) becomes

$$R(\mathbf{x}) = \sum_{r,s} P(\tau_r + \tau_s), \quad (12)$$

where we have neglected the amplitude factors and the summation is over all source-receiver pairs contributing to the image point.

Kirchhoff migration



Kirchhoff migration

Assume velocity $c(\mathbf{x})$ is constant, then

$$\begin{aligned}\tau_s &= \sqrt{\frac{(x - x_s)^2 + (y - y_s)^2}{c^2} + \tau_0^2}, \\ \tau_r &= \sqrt{\frac{(x - x_r)^2 + (y - y_r)^2}{c^2} + \tau_0^2},\end{aligned}\tag{13}$$

- ▶ $(x_r, y_r), (x_s, y_s)$: Source, receiver positions
- ▶ $\tau_0 = z/c$: vertical traveltime and
- ▶ c : velocity.

Kirchhoff migration

We then get for the total traveltime τ

$$\tau = \sqrt{\frac{(x - x_s)^2 + (y - y_s)^2}{c^2}} + \tau_0 + \sqrt{\frac{(x - x_r)^2 + (y - y_r)^2}{c^2}} + \tau_0. \quad (14)$$

The image R is then

$$R(x, y, \tau_0) = \sum_{r,s} P(\tau_s + \tau_s), \quad (15)$$

Zero-offset simple migration

We get for zero-offset (stack) data (2D)

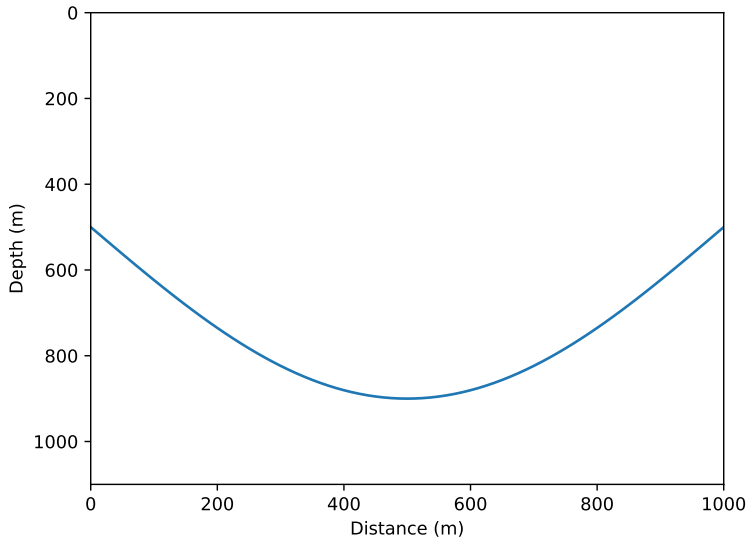
$$\tau = 2\sqrt{\frac{(x - x_s)^2}{c^2} + \tau_0^2}. \quad (16)$$

Zero-offset simple migration

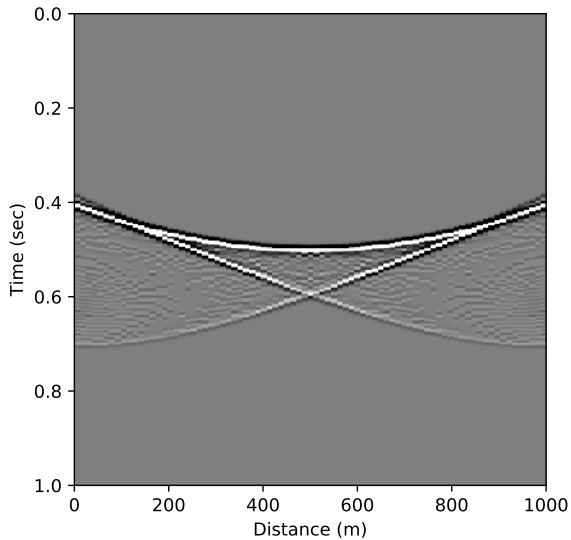
Migration of zero-offset section can then be done as:

- ▶ Chose a point x_s, y_s on the section.
- ▶ Loop over all x values, compute τ and sum all $P(\tau)$
(Summing over an hyperbola with center at x_s, τ_0 .)
- ▶ Put the sum at location x_s, y_s in the output.
- ▶ Repeat for all possible points x_s, y_s .

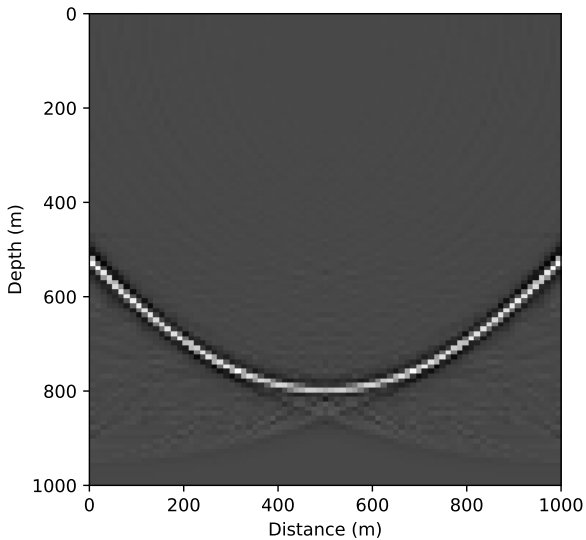
Syncline model



Syncline data



Syncline migration



Basic+ processing sequence

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